

Survey on Faults/Problems in NISQ Systems

To start, we will define precisely what we refer to when using the term Noisy Intermediate-Scale Quantum (NISQ) System. We consider any system composed of a quantum circuit inside a loop of classical optimization to be a NISQ system. It can be a Variational Quantum Algorithms (VQA), or any other kind of Hybrid Quantum – Classical system such as Quantum Approximate Optimization Algorithm (QAOA) or Quantum Machine Learning.

We consider a fault any wrong or suboptimal behaviour of a system: it can result to a crash, but also a wrong output, slow performance, or any other unexpected behaviour.

* Indique une question obligatoire

1. Your email (Optional)

2. What is your current job title? *

(For students, please, indicate which degree you are studying now)

3. What is your academic background? *

Une seule réponse possible.

☐ Physics

☐ Computer Science

☐ Autre :

4. What is your overall work experience? *

Une seule réponse possible.

- ☐ 0 to 1 year
- ☐ 1 to 3 years
- ☐ 3 to 10 years
- ☐ Over 10 years

5. What is your experience with Quantum Computing? *

Une seule réponse possible.

- ☐ 0 to 1 year
- ☐ 1 to 3 years
- ☐ 3 to 10 years
- ☐ Over 10 years

6. Have you encountered a NISQ fault/bug? *

Une seule réponse possible.

- ☐ Yes
- ☐ No

7. What type of NISQ Systems have you worked with? *

(For example VQE, QAOA, Quantum Machine Learning...)

8. What problem were you trying to tackle with NISQ Systems? *

Plusieurs réponses possibles.

- ☐ Quantum systems simulation
- ☐ Theoretical research
- ☐ QML Classifiers
- ☐ Autre : _____

9. Which Programming Languages and Frameworks have you been using? *

(Python, C++, Qiskit, PennyLane, Cirq...)

10. Have you encountered problems related to: *

If you answer yes: Please add an answer on the severity of the problem, and the difficulty to fix it.

Plusieurs réponses possibles.

	Yes	No	Minor Problem	Major Problem	Critical Problem	Low effort to fix	Medium effort to fix
Conceptualisation: For example, mapping the problem to a Hamiltonian, Breaking the electronic structure, Ansatz structure, Definition of the cost Function...	☐	☐	☐	☐	☐	☐	☐
Parametrisation: For example, in the tensors, operation pool, initialisation...	☐	☐	☐	☐	☐	☐	☐
Training: For example Iteration Problems, Storing intermediate results, Matrix calculation...	☐	☐	☐	☐	☐	☐	☐
Quantum circuit: For example in the circuit decomposition, Handling of gates and custom gates, operators, Ansatz creation...	☐	☐	☐	☐	☐	☐	☐
Measurement: For example Noisy measurements, in the expectation values, Precision, Observables, Memory use...	☐	☐	☐	☐	☐	☐	☐



11. Have you encountered problems related to: *

If you answer yes: Please add an answer on the severity of the problem, and the difficulty to fix it.

Plusieurs réponses possibles.

	Yes	No	Minor Problem	Major Problem	Critical Problem	Low effort to fix	Medium effort to fix	High effort to fix
API: For example, API Misuse, Depreciated API, Suboptimal API integration with other frameworks...	☐	☐	☐	☐	☐	☐	☐	☐
GPU usage: For example, Memory allocation, Memory Handling, Running on CPU instead of GPU...	☐	☐	☐	☐	☐	☐	☐	☐

12. Have you ever encountered any problems/bugs related to NISQ Systems that have not been mentioned in this survey? If yes, could you, please, describe them:

13. Any other comments

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